

Fermat's Little Theorem and fast modular exponentiation

(50 points) Calculate the modular multiplicative inverse for $e = 68$ for $GF(233)$, i.e. for arithmetic modulo 233, using Fermat's little theorem. You must show your work, and you must use fast modular exponentiation by repeated squaring (and show the work for that, too).

Multiplicative inverse is $68^{233-2} \pmod{233} = 24$

Note that on the exam you'll need to show your work. In particular, you need to show the steps of fast modular exponentiation *via* repeated squaring.

$$68^{-1} \equiv 68^{2^{33}-2} = 68^{2^{31}} \pmod{233}$$

$$231_{10} = 11100111_2$$

$$= 2^7 + 2^6 + 2^5 + 2^2 + 2^1 + 2^0$$

$$= 128 + 64 + 32 + 4 + 2 + 1$$

Repeated squaring:

$$68^2 = 4624 \equiv 197 \pmod{233}$$

$$68^4 \equiv 197^2 = 38809 \equiv 131 \pmod{233}$$

$$68^8 \equiv 131^2 = 17161 \equiv 152 \pmod{233}$$

$$68^{16} \equiv 152^2 = 23104 \equiv 37 \pmod{233}$$

$$68^{32} \equiv 37^2 = 1369 \equiv 204 \pmod{233}$$

$$68^{64} \equiv 204^2 = 41616 \equiv 142 \pmod{233}$$

$$68^{128} \equiv 142^2 = 20164 \equiv 126 \pmod{233}$$

Therefore:

$$68^{2^{31}} \equiv 126 \cdot 142 \cdot 204 \cdot 131 \cdot 197 \cdot 68$$

$$\equiv 184 \cdot 204 \cdot 131 \cdot 197 \cdot 68$$

$$\equiv 23 \cdot 131 \cdot 197 \cdot 68 \equiv 217 \cdot 197 \cdot 68$$

$$\equiv 110 \cdot 68 \equiv 24 \pmod{233}$$

Check: $68 \cdot 24 = 1632 = 7 \cdot 233 + 1$. So $68^{-1} \equiv 24 \pmod{233}$.